

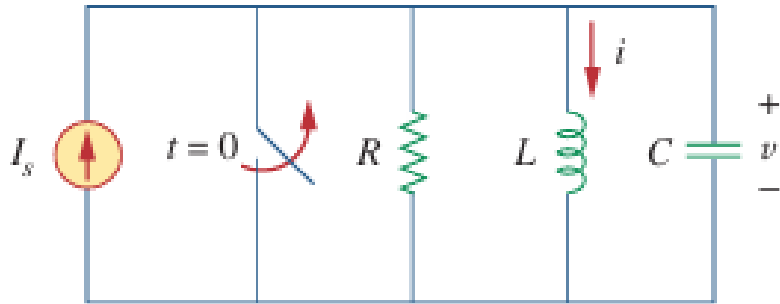
Module 2

Transient Analysis of First and Second-Order Circuits

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Step response of parallel RLC circuit



Applying KCL at the top node for $t > 0$

$$\frac{v}{R} + i + C \frac{dv}{dt} = I_s \quad \text{--- 1}$$

$$v = L \frac{di}{dt}$$

$$\frac{d^2 i}{dt^2} + \frac{1}{RC} \frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC} \quad \text{--- 2}$$

The complete solution to Eqn.2 consists of the transient response $i_t(t)$ and the steady-state response $i_{ss}(t)$

$$i(t) = i_t(t) + i_{ss}(t) \quad \text{--- 3}$$

Step response of parallel RLC circuit

$$i(t) = I_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad (\text{Overdamped})$$

$$i(t) = I_s + (A_1 + A_2 t) e^{-\alpha t} \quad (\text{Critically damped})$$

$$i(t) = I_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad (\text{Underdamped})$$

--- 4

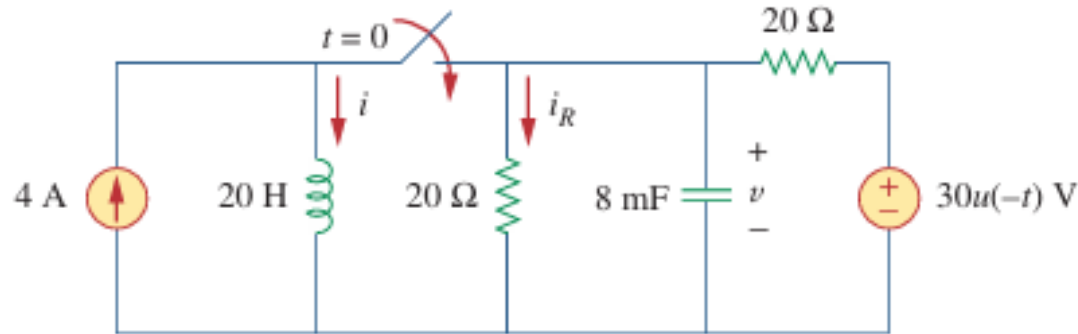
The constants and in each case can be determined from the initial conditions for i and di/dt .

The complete response for any variable may be found directly, using

$$x(t) = x_{ss}(t) + x_t(t)$$

Exercise

In the circuit, find $i(t)$ and $i_R(t)$ for $t > 0$.



For $t < 0$, the switch is open, and the circuit is partitioned into two independent sub-circuits.

The 4-A current flows through the inductor

$$i(0) = 4 \text{ A} \quad \text{--- 1}$$

$30 u(-t) = 30 \text{ V}$ when $t < 0$, and *equal to 0* when $t > 0$

$$v(0) = \frac{20}{20 + 20}(30) = 15 \text{ V} \quad \text{--- 2}$$

Exercise

For $t > 0$, the switch is closed, and we have a parallel RLC circuit with a current source. The voltage source is zero which means it acts like a short-circuit. The two resistors are now in parallel.

$$\begin{aligned}\alpha &= \frac{1}{2RC} = \frac{1}{2 \times 10 \times 8 \times 10^{-3}} = 6.25 \\ \omega_0 &= \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{20 \times 8 \times 10^{-3}}} = 2.5 \\ s_{1,2} &= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -6.25 \pm \sqrt{39.0625 - 6.25} \\ &= -6.25 \pm 5.7282 \\ s_1 &= -11.978, \quad s_2 = -0.5218 \quad \text{--- 3}\end{aligned}$$

Since $\alpha > \omega_0$ we have the over damped case.

$$i(t) = I_s + A_1 e^{-11.978t} + A_2 e^{-0.5218t} \quad \text{--- 4}$$

I_s the final value of $i(t)$ which is equal to 4 A .

Use the initial conditions to determine A_1 and A_2 .

Exercise

$$i(0) = 4 = 4 + A_1 + A_2 \quad \Rightarrow \quad A_2 = -A_1$$

$$\frac{di}{dt} = -11.978A_1e^{-11.978t} - 0.5218A_2e^{-0.5218t}$$

At $t = 0$

$$\frac{di(0)}{dt} = -11.978A_1 - 0.5218A_2 \quad \text{--- 5}$$

But,

$$L \frac{di(0)}{dt} = v(0) = 15 \quad \Rightarrow \quad \frac{di(0)}{dt} = \frac{15}{L} = \frac{15}{20} = 0.75$$

Applying this in Eqn.5, we get

$$0.75 = (11.978 - 0.5218)A_2 \quad \Rightarrow \quad A_2 = 0.0655$$

Thus, $A_1 = -0.0655$ and $A_2 = 0.0655$

$$i(t) = 4 + 0.0655(e^{-0.5218t} - e^{-11.978t}) \text{ A}$$

$$i_R(t) = \frac{v(t)}{20} = \frac{L}{20} \frac{di}{dt} = 0.785e^{-11.978t} - 0.0342e^{-0.5218t} \text{ A}$$

Next class: General Second order circuits